Weekly Lesson Plan – Week 1 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: August 10, 2026 | Course: Principles of Applied Engineering

TEKS / ELPS:

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TEKS: §127.15(d)(1) CTE employability skills - professional standards; 127.391(d)(2)(B) maintain a design and computation engineering notebook; §127.391(d)(1)(A) history of engineering disciplines; (d)(1)(E) compare engineering career paths; (d)(2)(C) sketching and CADD to present ideas; §127.391 engineering design process - define a problem, generate ideas, plan a solution; §127.391 develop & refine a design; use digital tools to model & communicate it; §127.391 present, evaluate & give feedback on a design solution
ELPS: c.1A, c.2D, c.3C, c.4G, c.5B
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Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will...*

MONDAY AUG 10	TUESDAY AUG 11	WEDNESDAY AUG 12	THURSDAY AUG 13	FRIDAY AUG 14
Students will set up a standards-compliant engineering design notebook (cover, table of contents, numbered and dated pages) and write a complete first entry with a labeled sketch and measurements recorded with correct units and tolerance, then evaluate a sample entry against a professional-documentation rubric.	Students will distinguish engineering from science, identify at least four engineering disciplines with a historical milestone, map an engineered product's subsystems to the disciplines and career paths behind them, and produce an annotated technical sketch (isometric or orthographic) with labeled subsystems and dimensioned callouts.	Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan).	Students will finish their final design, write a structured AI prompt, and generate + submit a product image.	Students will present their product - communicating the problem it solves, their final design, and how it works - and give supportive peer feedback.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.

Assessment of Content Objective: *Instrument used to determine mastery. Ex: Exit Ticket.*

Students will be assessed by written, group, or oral responses throughout the class period.

Set (Bell Work): *A motivating action that engages students in the objective – question, fact, activity. Essential Question:*

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You invent something valuable and a competitor copies it. Was that wrong? Why?	List 5 engineered objects within reach right now. Next to each, name the problem it solves.	What is the purpose of the engineering design cycle?	Watch: how a product improves, version to version	Write your script before you present

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...*

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Step 1: Bell ringer / essential question. Step 2: Bell-work: You invent something valuable; a competitor says they thought of it first. What evidence would prove it was you? (write in notebook) (7 min) Step 3: Why engineers document: IP/legal, reproducibility, traceability, quality & safety (notes + notebook standards) (10 min) Step 4: Anatomy of an entry + units/tolerance, modeled live under the doc cam (8 min) Step 5: Set up notebook (cover, TOC, page numbers) + write a complete first entry with a labeled, dimensioned sketch (12 min) Step 6: Rubric self-audit + Google Classroom sign-in (5 min) Step 7: Exit ticket: 3 features that make a notebook legally defensible (3 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: list 5 engineered objects within reach; next to each, guess which kind of engineer made it (6 min) Step 3: Science vs engineering + design under constraints (criteria/constraints/trade-offs) (8 min) Step 4: Watch: Crash Course Engineering #1 + guided viewing (disciplines & history) (12 min) Step 5: History of the disciplines + careers, pay, and pathways (8 min) Step 6: Technical sketching mini-lesson + annotate an engineered object (8 min) Step 7: Exit ticket (3 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: purpose of the engineering design cycle (5 min) Step 3: Present: the 5-step engineering design process (16 min) Step 4: Pick a real problem (list 3, choose 1) (7 min) Step 5: Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials) (17 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work + video (start 1:28): name 2 improvements and why each matters (8 min) Step 3: Model the notebook page (prompt on top, sketch on bottom) + copy the prompt (10 min) Step 4: Finish final design, then generate the product image in Gemini (20 min) Step 5: Submit the image to Google Classroom (7 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: write your 4-sentence presentation script (5 min) Step 3: Model: the four things every presenter says + how to be a great audience (8 min) Step 4: Presentations: each student shows their AI product image + notebook page (~1 min each) (32 min) Step 5: Exit ticket: one thing you would design next (5 min)

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state what they accomplished and what worked best.
Study Skills: *Skills implemented by students – note-taking, summarizing, teamwork, etc.*
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.
SPED & 504 Accommodations / Modifications: *Codes.*
Per Individualized Education Plan (IEP) / 504 plan.
ELL Accommodations: *Codes.*
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).